Do Now · retrieval

Dijkstra: shortest paths I

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Why does quicksort exclude the pivot from its recursive calls?

Q2

Define a weighted edge.

Q3

What is the first index of an array?

Q4

A–B costs 3 and B–C costs 5. What is cost of A→B→C?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Initialise from the start — Distance(A)=0, all others infinity. Relax A's edges to set B=4 and C=2.

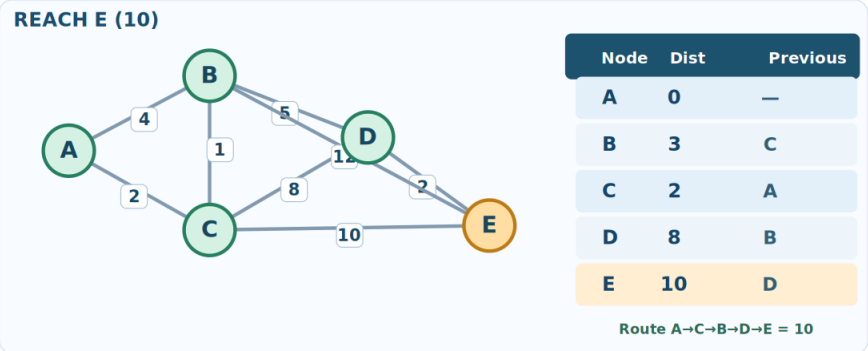
Core idea 2

Always choose smallest tentative distance — C=2 settles before B=4. Through C, B improves to 3.

Core idea 3

Keep updating neighbours — Settling B changes D from 10 to 8; settling D changes E from 12 to 10.

Key visual · complete concept



New Learning · applying the ideas

Core idea 4

Reconstruct from predecessors — E←D←B←C←A. Reverse these arrows for A→C→B→D→E, total distance 10.

Core idea 5

Dijkstra's requirement — Edge weights must be non-negative. A visited/settled node's shortest distance is then final.

Vocabulary · use the correct technical terms

tentative distance — Best known cost to an unsettled vertex.

settled — Vertex whose shortest distance has been finalised.

relaxation — Improve a neighbour's distance via the current vertex.

predecessor — Previous vertex on the best-known route.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

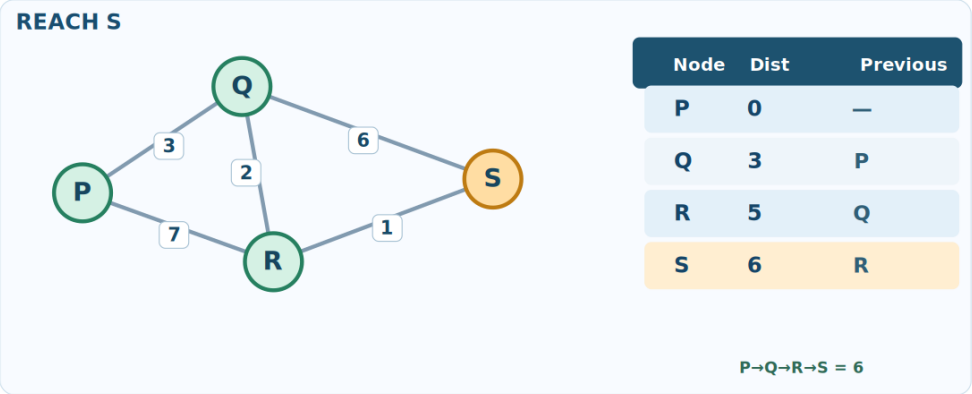
Problem

Find the shortest P→S route using non-negative weights.

Follow the worked method

1. Predict the next state — P gives tentative Q=3 and R=7.
2. Reveal and explain — Settle R=5, improving S to 6.
3. Reveal and explain — Predecessors S←R←Q←P give P→Q→R→S, cost 6.

Source model · final illustrated step



Node	Dist	Previous
P	0	—
Q	3	P
R	5	Q
S	6	R

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict the next state
From M, set N=5 and O=2; O is the next vertex settled.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer every question in BLACK PEN. Show steps and use OCR command words.

Question 1

A–B=4 and A–C=2. Starting Dijkstra at A, state the tentative distances of A, B and C after relaxing A's neighbours.

Question 2

On edges S–A=3, S–B=7, A–B=2, B–T=2 and A–T=8, find the shortest S→T path with full distance reasoning.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer every question in BLACK PEN. Show steps and use OCR command words.

Question 3

Explain why Dijkstra must not be relied upon when negative-weight edges are possible, and contrast tentative and settled distances.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Dijkstra always chooses the next vertex by its smallest edge weight.

Use key vocabulary from memory

1. Explain tentative distance in your own words.

2. Explain settled in your own words.

3. Explain relaxation in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.